

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Vision + Text (Tri-modal VLM)
- TS + Knowledge Graph / Topology
 - TS + Audio / Waveforms
- Omnimodal / Multi-sensor Data

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, GPT4TS)
 - Visual Rendering (VisionTS)
- Unified Early Tokenization (UniTS, ChatTS)
 - Cross-Attention Adapter (Time-VLM)
- Contrastive Dual-Tower (TRACE, TEST)
- Prompt Serialization (PromptCast, LLMTIME)

3. Non-TS Role

- Auxiliary Context / Condition
 - Frozen Transfer Substrate
- Conversational & Reasoning Interface
- Metric Alignment Target (Retrieval)
- Multi-Task Instruction Steering
- Tool & Neuro-Symbolic Querying

4. Downstream Tasks

- Multimodal Forecasting (Point/Prob.)
- Anomaly Detection & Explanation
 - Time Series QA & Reasoning
- Cross-Modal Retrieval (TS <-> Text)
- Automated Report / Caption Gen
- Zero-Shot & Few-Shot Generalization

Application Domains: Healthcare & EHR • Finance & Markets • Energy & Smart Grids • Meteorology & Climate • Traffic & Smart Cities